

Multiple Choice (10 marks)

1. Which of the following is NOT a group?
 - (a) The integers under addition
 - (b) The nonzero integers under multiplication
 - (c) The nonzero real numbers under multiplication
 - (d) The complex numbers under addition
2. Using Fermat's theorem, find the remainder of 3^{47} when it is divided by 23.
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. If a real number x is chosen at random in the interval $[0, 3]$ and a real number y is chosen at random in the interval $[0, 4]$, what is the probability that $x < y$?
 - (a) $1/2$
 - (b) $7/12$
 - (c) $5/8$
 - (d) $2/3$
4. Statement 1 | The external direct product of cyclic groups is cyclic. Statement 2 | The external direct product of D_3 and D_4 is isomorphic to D_{12} .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. There are 25 suitcases, 5 of which are damaged. Three suitcases are selected at random. What is the probability that exactly 2 are damaged?
 - (a) $2/69$
 - (b) $1/30$
 - (c) $2/23$
 - (d) $12/125$

True / False (10 marks)

Answer True or False

6. True or False: There exists a non-abelian group whose subgroup is non-abelian but still considered abelian.
7. True or False: The element $c = 2$ in $\mathbb{Z}_3$ makes the polynomial $x^3 + x^2 + c$ irreducible, thus $\mathbb{Z}_3[x]/(x^3 + x^2 + c)$ is a field.
8. True or False: The relation $S = \{(1, 1), (2, 2)\}$ is an equivalence relation.
9. True or False: Every element of a group indeed generates a cyclic subgroup of the group.
10. True or False: There is only one additive abelian group of order 16 where $x + x + x + x = 0$ for every element x in the group.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Each letter represents a non-zero digit. What is the value of t ?

$$c + o\& = u$$

$$u + n\& = t$$

$$t + c\& = s$$

$$o + n + s\& = 12$$

12. Given a statement $S(n)$ about positive integers n , where $S(k)$ implies $S(k + 1)$ and $S(n_0)$ is false for some positive integer n_0 , discuss the implications for $S(n)$ for $n \leq n_0$.

End of Paper